

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

The R2 S0/0/1 interface is currently in a passive state and is not transmitting OSPF Hello packets; therefore, R3 cannot recognise it as a neighbor.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65. The way to compute is: R3 = R2 S0/0/1 cost(64) + R2 = G0/0 cost(1).

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Because the default reference bandwidth is 100Mb/s, the cost of all links faster than 100Mb/s (such as Gigabit Ethernet or 10G Ethernet) will become 1. Increasing this value allows OSPF to determine the true cost difference on higher-speed links.

Step 2: Change the bandwidth for an interface.

- g. Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

To 192.168.3.0/24: R1 S0/0/1 cost + R3 G0/0 cost (781+1 = 782).

To 192.168.23.0/30: R1 S0/0/1 cost + R3 S0/0/1 cost (781+ 64= 845).

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562. Since it will take two serial links (e.g. via either R2 or R3) to get to the destination, the cost of each Serial link with bandwidth 128 set is 781, so 781 + 781 = 1562.

Step 3: Change the route cost.

- c. Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The direct link (S0/0/1) from R1 to R3 had was manually set by hand to be 1565 in cost. If the detour through R2, the cumulative cost would be only 1563 (R1 S0/0/0 is 781 + R2 S0/0/1 is 781+ R3 G0/0 is 1). OSPF will always choose the path that has the least amount of cost.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

To keep OSPF stable as a routing domain. If the Router ID depends on the physical interface, the Router ID will recalculate and change when the interface is down. To prevent this, use the Loopback interface or the router-id command.

2. Why is the DR/BDR election process not a concern in this lab?

This is because Routers are attached via Point-to-Point serial links, and DR/BDR Elections only occur in Multi-access Networks (e.g. Ethernet Broadcast)

3. Why would you want to set an OSPF interface to passive?

This prevents OSPF routing update packets from being sent to networks that will not require OSPF routing information (e.g., a local area network connected to a PC), reducing unnecessary traffic and saving bandwidth.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF Areas

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0.

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

EIGRP. The router will look at Administrative Distance (AD) instead of the metric when comparing routing protocols. EIGRP's default administrative distance is 90, which is less than OSPF's 110 and RIP's 120; smaller values are preferred.